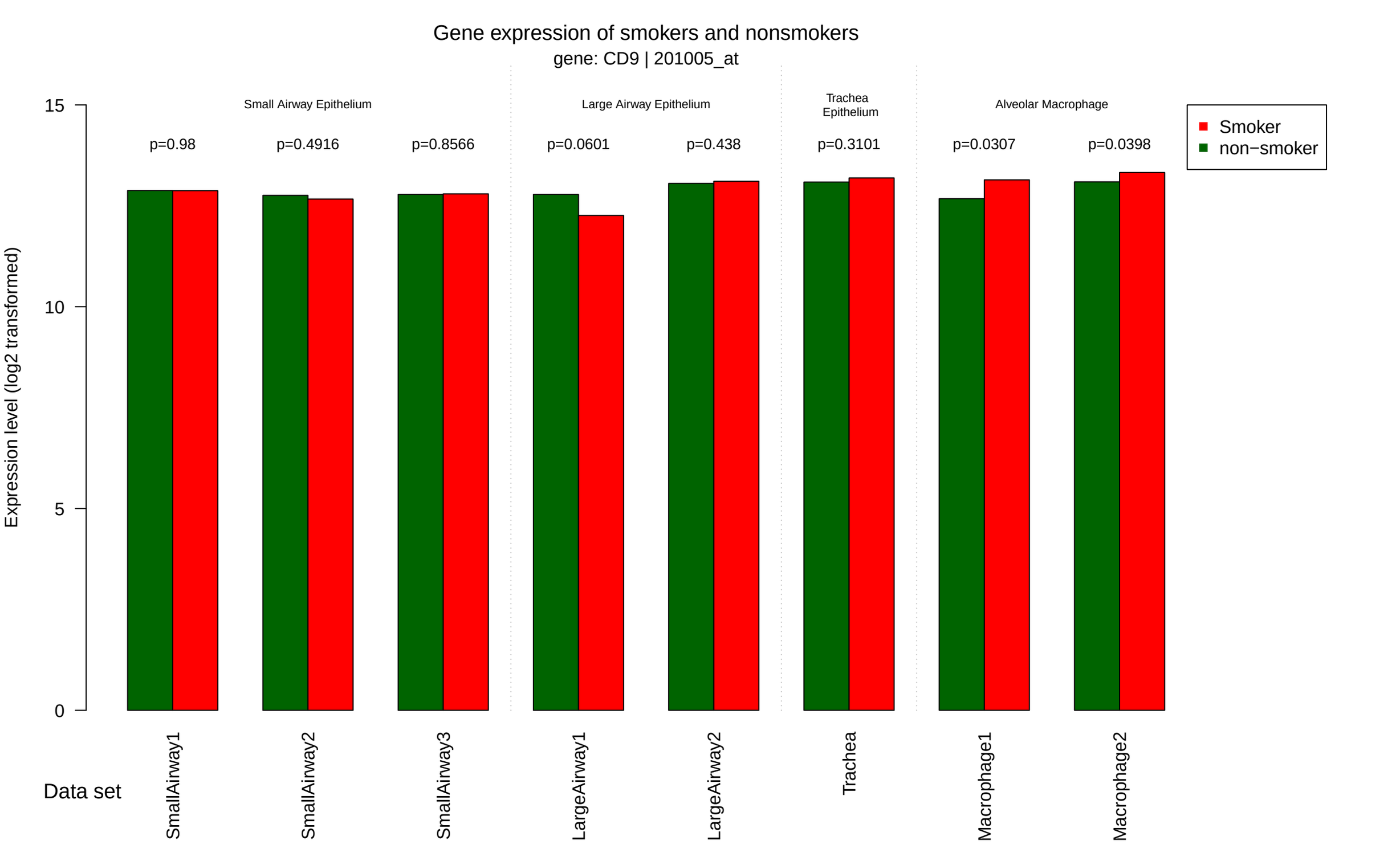
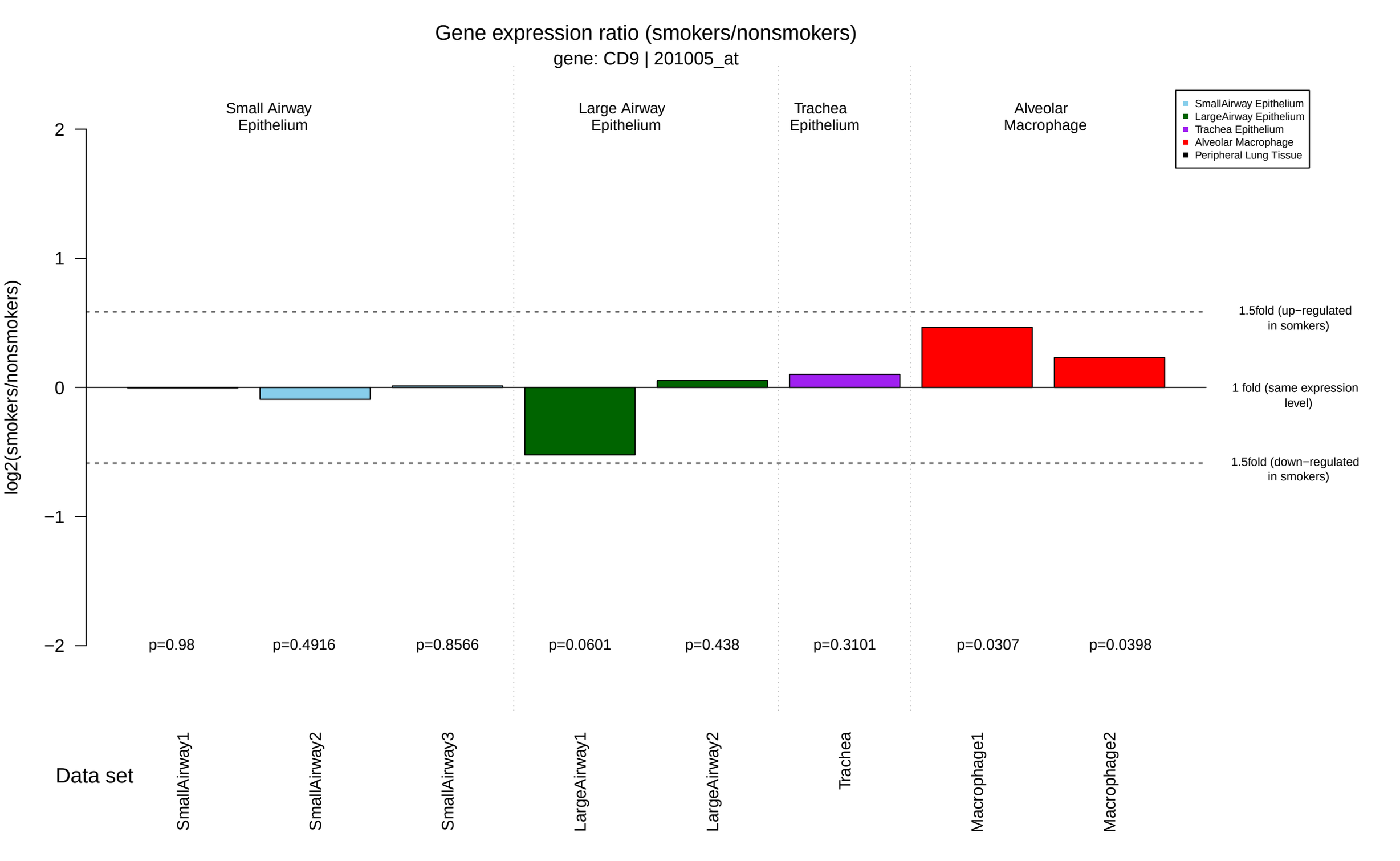
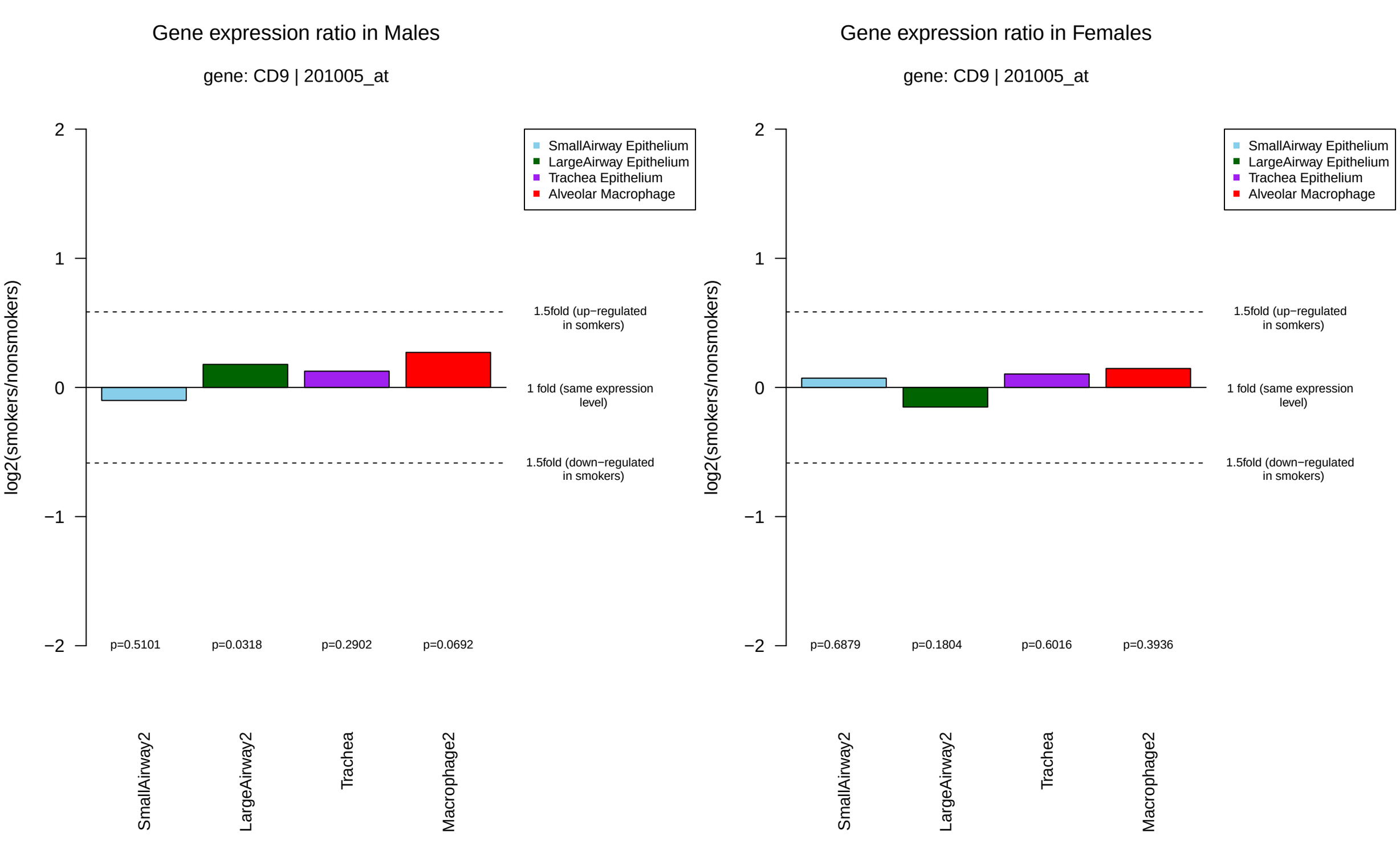
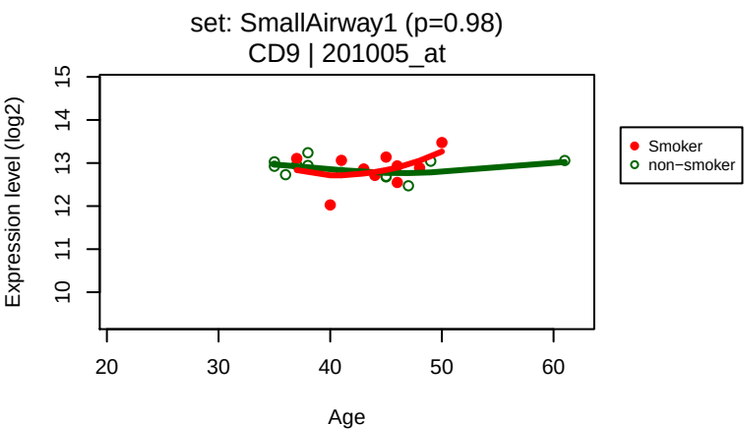
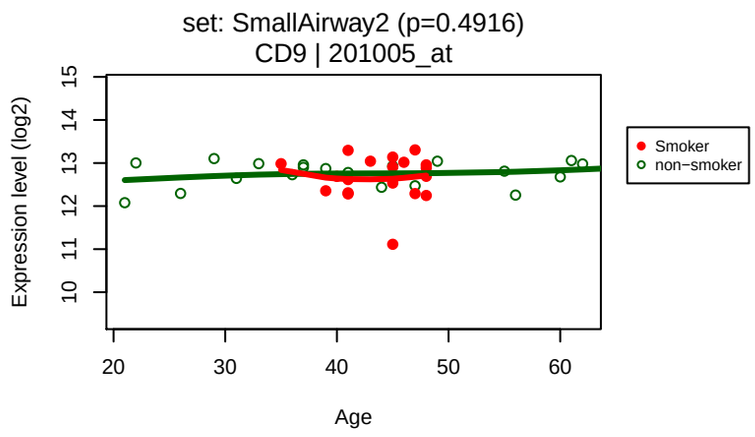
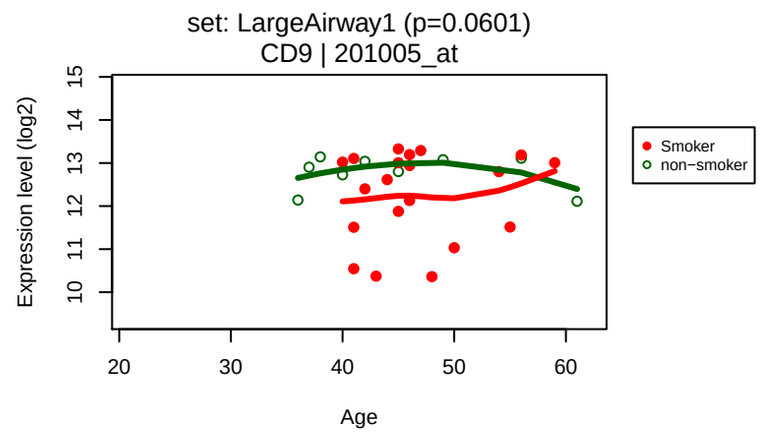

Gene expression of smokers and nonsmokers
gene: CD9 | 201005_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.98
p=0.4916
p=0.8566
p=0.0601
p=0.438
p=0.3101
p=0.0307
p=0.0398
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

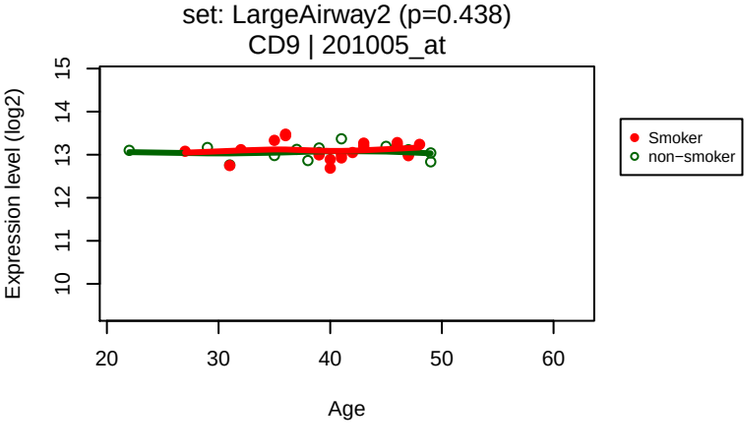

Gene expression ratio (smokers/nonsmokers)
gene: CD9 | 201005_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.98
p=0.4916
p=0.8566
p=0.0601
p=0.438
p=0.3101
p=0.0307
p=0.0398
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

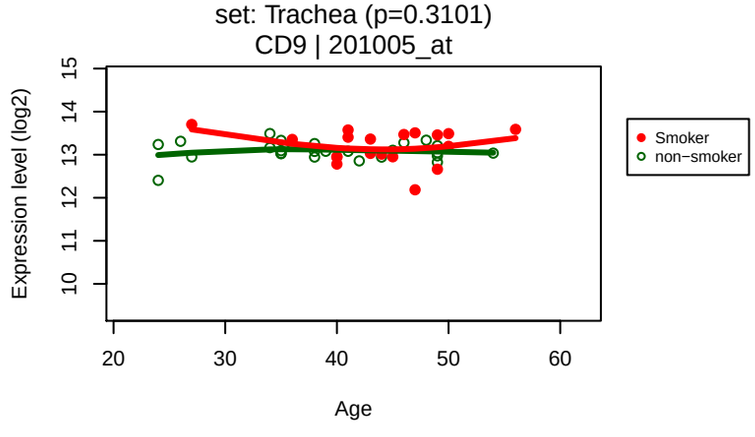

Gene expression ratio in Males
gene: CD9 | 201005_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5101
p=0.0318
p=0.2902
p=0.0692
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CD9 | 201005_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6879
p=0.1804
p=0.6016
p=0.3936
SmallAirway2
LargeAirway2
Trachea
Macrophage2

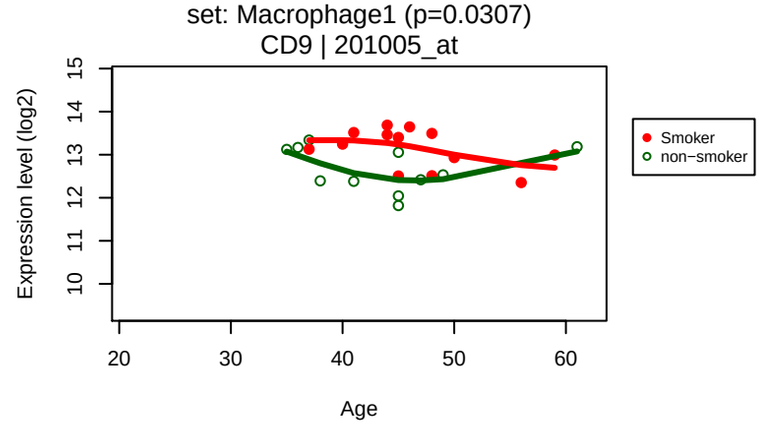

set: SmallAirway1 (p=0.98)
CD9 | 201005_at
Expression level (log2)
Age
Smoker
non−smoker

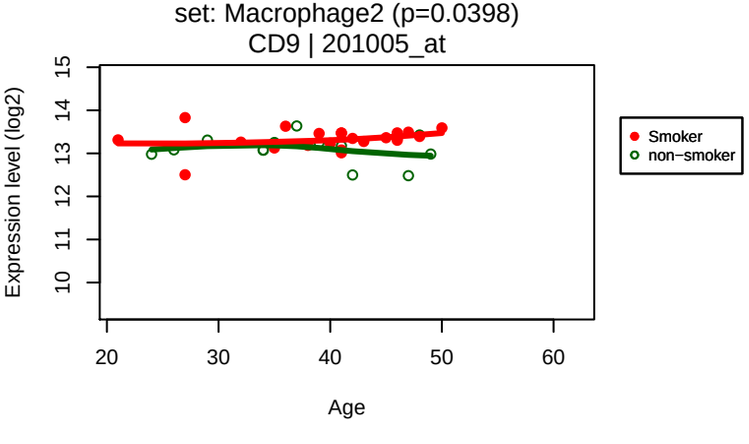

set: SmallAirway2 (p=0.4916)
CD9 | 201005_at
Expression level (log2)
Age
Smoker
non−smoker

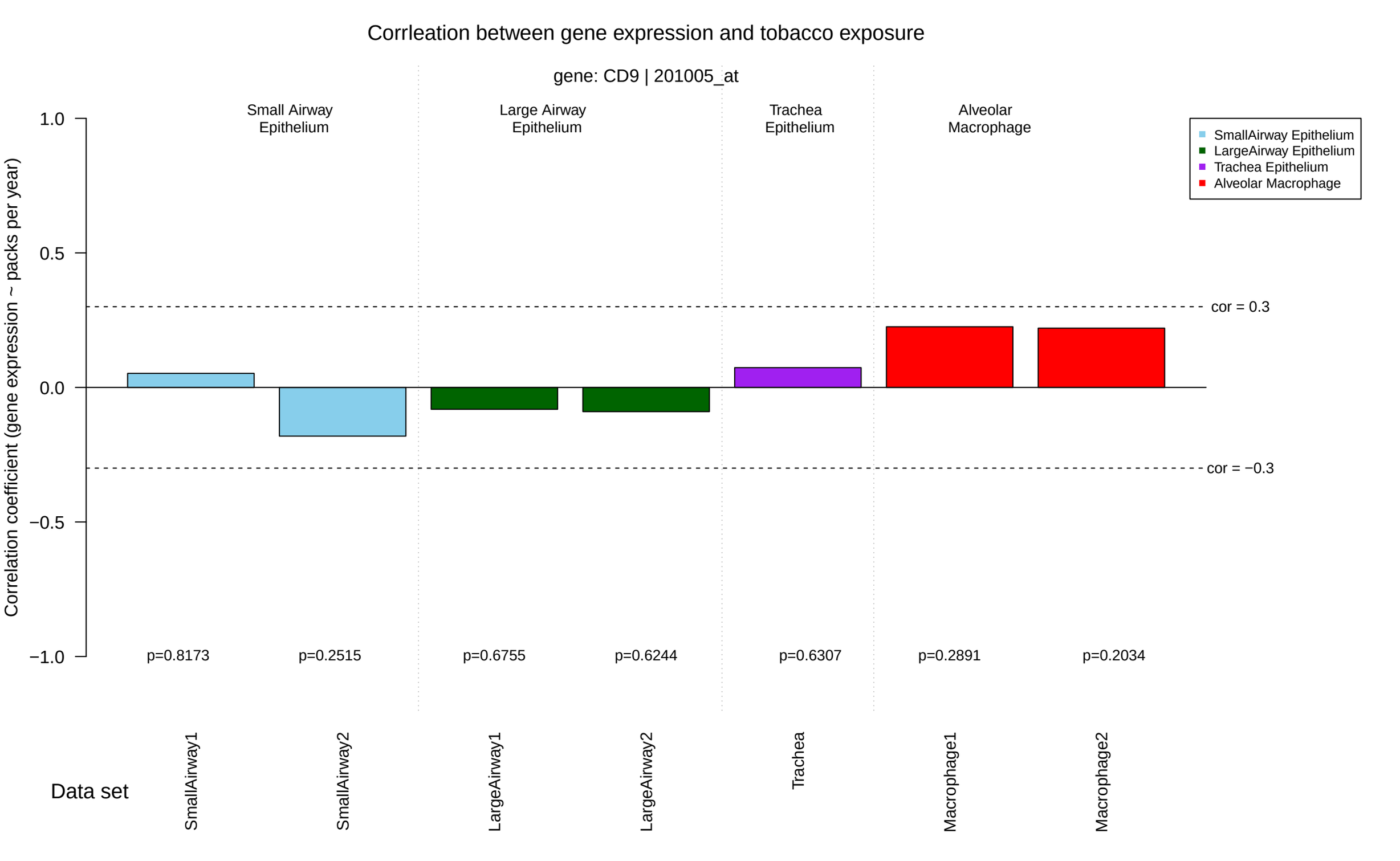

set: LargeAirway1 (p=0.0601)
CD9 | 201005_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.438)
CD9 | 201005_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3101)
CD9 | 201005_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0307)
CD9 | 201005_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0398)
CD9 | 201005_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CD9 | 201005_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8173
p=0.2515
p=0.6755
p=0.6244
p=0.6307
p=0.2891
p=0.2034
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2